

# SHAH NAWAZ HAIDER

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## EDUCATION

### University of Science and Technology Chittagong (USTC)

Chittagong, Bangladesh

B.Sc. in Computer Science & Engineering. CGPA: 3.72/4.00

May 2026

**Thesis:** QuantFlow: A Federated Mamba-Based Post-Transformer Foundation Model for Time-Series Forecasting

**Advisors:** Dr. Hadaate Ullah & Prof. Sarowar Morshed Shawon

## RESEARCH INTERESTS

Foundation Models, World Models, Large Language Models, Agentic Reasoning, Multimodal Learning, Interpretability

## HIGHLIGHTS

- 13 peer-reviewed / preprint publications, including 2 as first author.
- Research Lead at ResearchBuddy AI, directing two applied NLP research teams.
- Peer Reviewer for high-impact publishers, including Elsevier and Springer Nature.
- Industry experience building production LLM and ML systems for financial risk modeling at scale.

## RESEARCH & INDUSTRY EXPERIENCE

### Synnax AI

Dubai, UAE (Remote)

Data Scientist

Aug 2024 – May 2026

- Engineered end-to-end ML pipelines for credit risk and default prediction over SEC filings and market data spanning 16,000+ public companies.
- Built LLM-powered agents to extract risk signals from 10-K, 10-Q, and 8-K filings, automating detection of high-risk and defaulted entities.
- Converted large-scale unstructured regulatory data into production-ready features within a production MLOps stack.

### ResearchBuddy AI (Private Research Lab)

Research Lead

Jul 2026 – Present

- Lead two research teams focused on Bengali Dialect Corpus Enhancement and Edge-Efficient Embedding Models.
- Direct research strategy, oversee benchmarking, and coordinate publication efforts across both teams.

### Quantropy

Chittagong, Bangladesh

AI/ML Instructor

Jun 2026 – Present

- Designed and taught an AI curriculum covering deep learning, LLMs, foundation models, and modern generative AI.
- Ran hands-on workshops on transformers, retrieval-augmented generation (RAG), and AI agents.

### Omdena

Milan, Italy (Remote)

Data Scientist (Part-time)

May 2024 – Jul 2024

- Built and evaluated ML models for crop optimization and resource allocation from environmental and spatial agricultural datasets.

### University of Science and Technology Chittagong (USTC)

Chittagong, Bangladesh

Peer Tutor (Part-time)

Sep 2023 – Nov 2023

- Mentored undergraduates in programming, algorithms, and problem-solving within introductory computer science courses.

## SELECTED PUBLICATIONS

### First-Author

[1] **Haider, S. N.**, Austin, S., Barua, A., Shawon, S. M., & Ullah, H. QuantFlow: A Federated Mamba-Based Post-Transformer Foundation Model for Time-Series Forecasting. *arXiv:2607.02632* (Preprint).

[2] **Haider, S. N.**, Barua, A., Austin, S., & Shawon, S. M. Lightweight Federated and Explainable AI-Enhanced Crop Yield Forecasting in Heterogeneous Agricultural Systems. *IEEE ICCIT*, 2025. [DOI](#)

### Selected Co-Authored

[1] Shawon, S. M., **Haider, S. N.**, Barua, A., Austin, S., Adan, I. A., Hossain, M. S., & Zubair, H. T. Hybrid CNN-LSTM Model for Urban Energy Load Forecasting with IGA-XAI for Smart Grids. *Results in Engineering*, 2025. [DOI](#)

[2] Barua, A., **Haider, S. N.**, Austin, S., & Shawon, S. M. Urban Energy Load Dataset of Chattogram: Peak and Off-Peak Variability Insights. *Data in Brief*, 2025. [DOI](#)

[3] Shawon, S. M., Dutta, P., **Haider, S. N.**, Barua, A., Austin, S., & Adan, I. A. Agri-Light-Fed: Lightweight Federated Learning with Explainable AI for Efficient and Privacy-Preserving Crop Yield Prediction. *IEEE ICCIT*, 2025. [DOI](#)

[4] Shawon, S. M., **Haider, S. N.**, Barua, A., Austin, S., & Noor, K. R. Predictive Modelling for Multi-Location Deep Learning-Based Load Forecasting: An Integrated Approach. *IEEE ICECE*, 2024. [DOI](#)

[5] Austin, S., **Haider, S. N.**, Shil, R., Lija, N. S., & Noor, K. R. Bi-LSTM-RF Based Hybrid Deep Learning Approach to Identify Level of Depression from Bengali Facebook Status. *IEEE ICCIT*, 2024. [DOI](#)

[6] Shawon, S. M., **Haider, S. N.**, Chakma, A., Alam, M. W., Islam, M. T., & Rana, M. F. DeepFlowNet: Deep Learning-Based Daily Water Flow Forecasting of Test River. *IEEE PEEIACON*, 2024. [DOI](#)

[7] Shawon, S. M., **Haider, S. N.**, Austin, S., & Barua, A. HybridNet: ResNet50-GRU Integration for Enhanced Rice Leaf Disease Classification in Precision Agriculture. *Springer LNNS (MIET)*, 2024. [DOI](#)

Complete publication list and citation metrics available on [Google Scholar](#).

RESEARCH PROJECTS

**Qwen-NL2FOL: Natural Language to First-Order Logic Translation**

[GitHub](#)

- Distilled step-by-step NL-to-FOL reasoning traces from a large teacher model to build a custom training corpus.
- LoRA fine-tuned a 2B-parameter Qwen student to reason before translating natural language into First-Order Logic.
- Applied GRPO with a custom logical-equivalence reward, matching 13B-parameter LogicLLaMA accuracy on MALLS.

**Bengali BPE Tokenizer**

[GitHub](#)

- Trained a Byte Pair Encoding tokenizer from scratch for Bengali and other non-Latin scripts, achieving  $7.33\times$  fewer tokens than GPT-2 across 3.4 billion characters of the CC-100 Bengali corpus.
- Implemented custom multi-byte UTF-8 and conjunct handling, cutting token inflation and inference cost for under-resourced languages.

**Lunar Seismic Data Visualization - NASA International Space Apps Challenge**

[GitHub](#)

- Built an interactive web app to visualize lunar seismic data from NASA's Planetary Data System, overlaying mineral composition and Bouguer gravity disturbances against seismic events.

PROFESSIONAL SERVICE

- Peer Reviewer, *Array*, Elsevier
- Peer Reviewer, *SN Computer Science*, Springer Nature
- Peer Reviewer, *Applied Food Research*, Elsevier
- Peer Reviewer, *Frontiers in Big Data*, Frontiers

AWARDS

- |                                                                                                |               |
|------------------------------------------------------------------------------------------------|---------------|
| • <b>Champion, Project Competition</b> – Research Colloquium, USTC. Chittagong, Bangladesh     | May 2026      |
| • <b>Champion, Poster Competition</b> – Winter School Colloquium, USTC. Chittagong, Bangladesh | February 2025 |
| • <b>Regional Champion</b> – NASA International Space Apps Challenge. Dhaka, Bangladesh        | October 2023  |

TECHNICAL SKILLS

|                  |                                                                               |
|------------------|-------------------------------------------------------------------------------|
| Programming      | Python, Java, JavaScript, C/C++, MATLAB, SQL                                  |
| ML / DL          | PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers  |
| LLM & Agentic AI | LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, LoRA/QLoRA |
| Data & Databases | Pandas, NumPy, Matplotlib, Plotly, PostgreSQL, MongoDB, Redis                 |

LEADERSHIP & SERVICE

**Programming Club, USTC**

*President*

Chittagong, Bangladesh

Jul 2024 – Jun 2025

- Led the university programming club, organizing technical workshops and coding competitions for CS students.

**Robotics Society, USTC**

*Member*

Chittagong, Bangladesh

Jul 2023 – Jun 2025

- Architected an autonomous path-following robotic system by optimizing sensor fusion algorithms.

CERTIFICATIONS

- |                                                                           |          |
|---------------------------------------------------------------------------|----------|
| • <b>Machine Learning Specialization</b> , Stanford University (Coursera) | Dec 2023 |
| • <b>Data Analysis with Python</b> , freeCodeCamp                         | Mar 2023 |